

Artificial Intelligence

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Assistant Professor

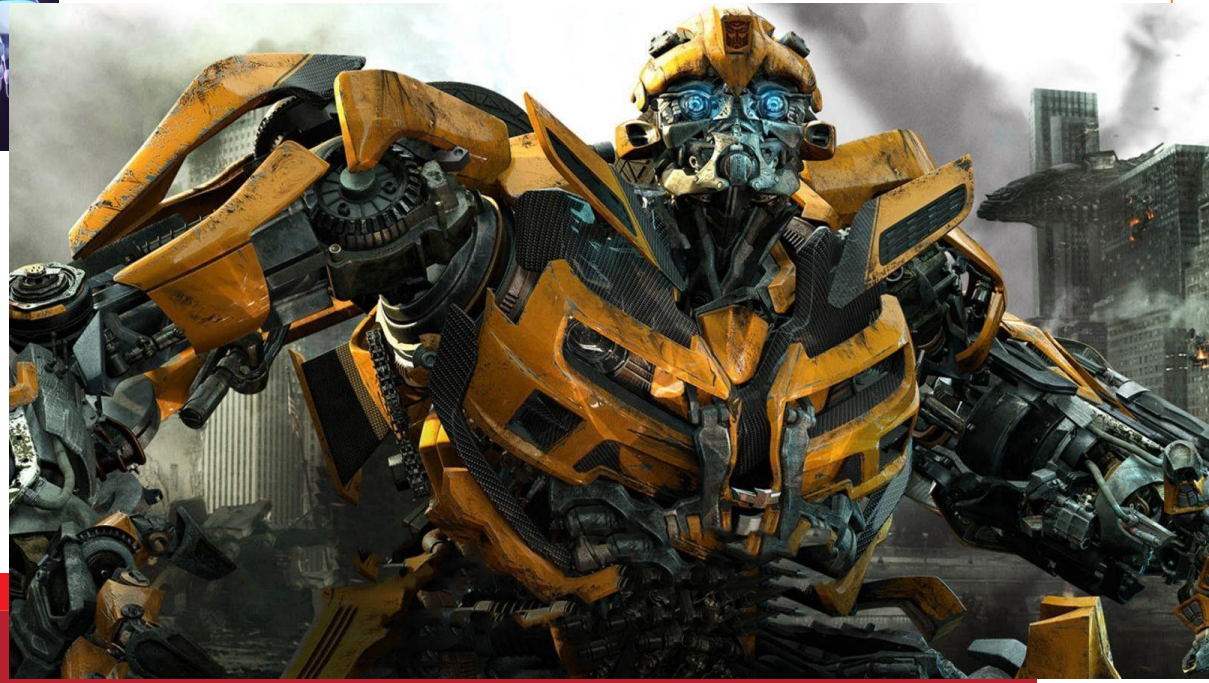
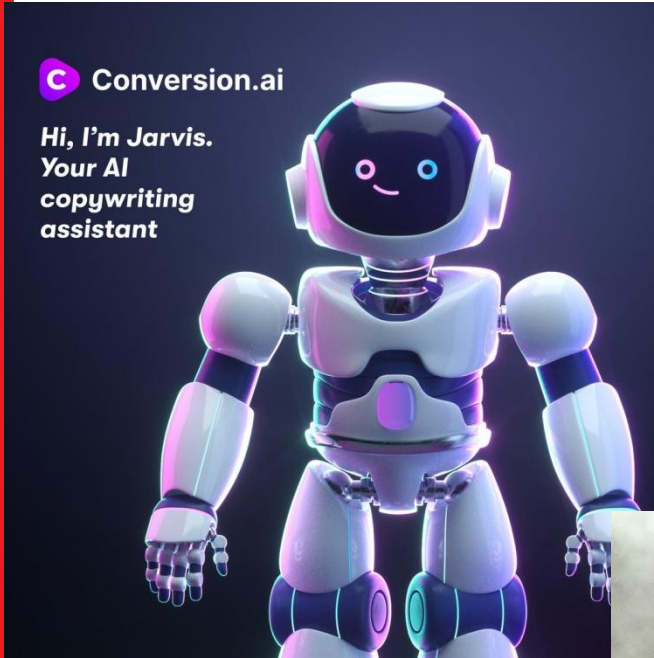
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Where have you seen AI before?



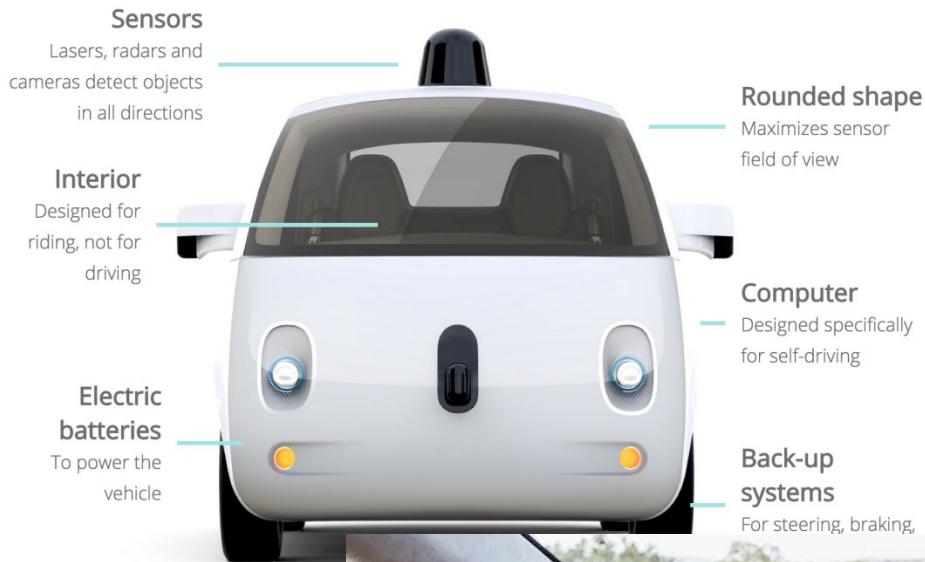
Where have you seen AI before?



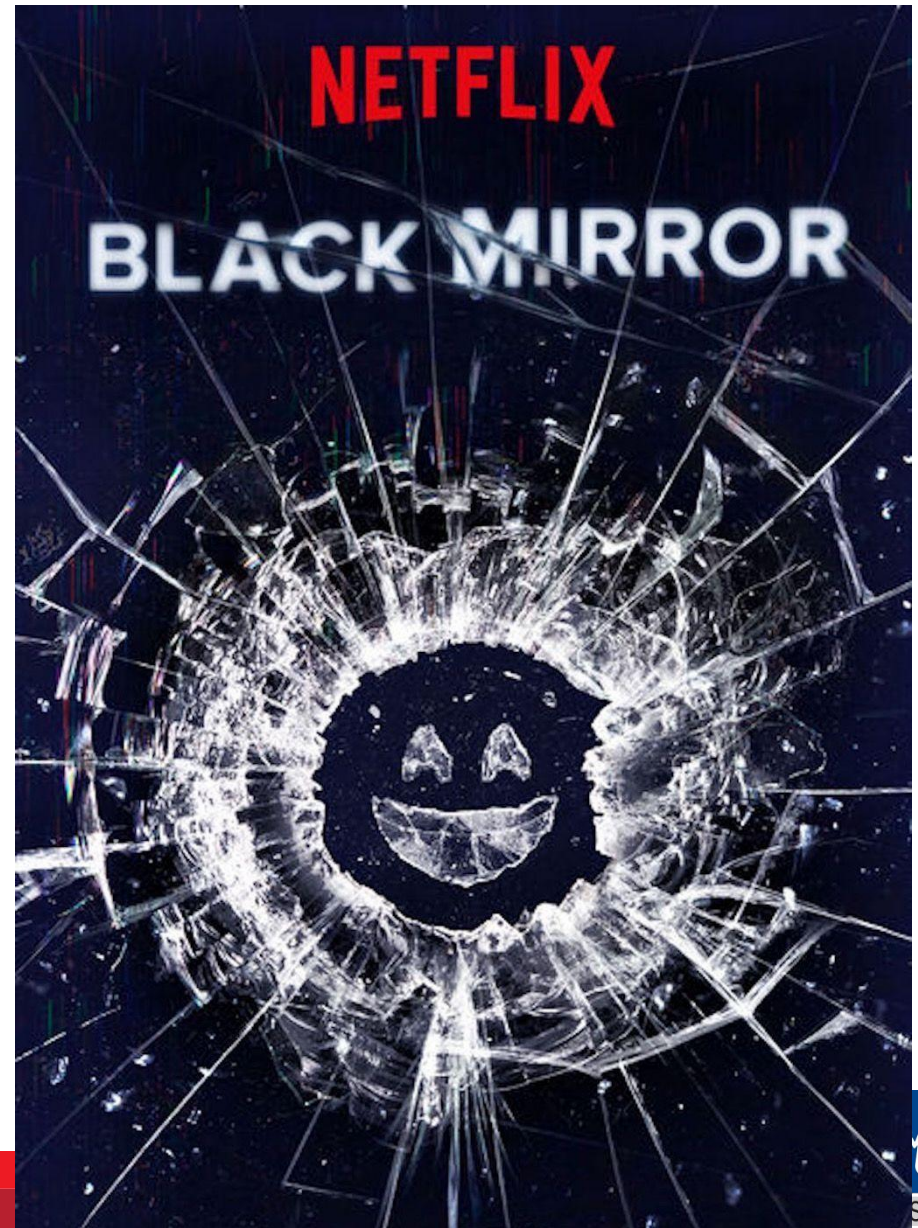
Where have you seen AI before?



Where have you seen AI before?



Where have you seen AI before?



WHAT YOU'LL STUDY IN AI

- AI– Toy problems, games
- Intelligent agents
- Algorithms- searching
- Knowledge representation & reasoning
- Learning
- Planning
- Expert systems

APPLICATIONS OF AI

- Brain simulation
- Cognitive architectures
- Games
- Knowledge and reasoning
- Motion and manipulation
- Natural language processing
- Planning

FUTURE IN THE FIELD OF AI

- Planning
- Knowledge Engineering
- Information retrieval
- Game Theory
- Expert systems
- Natural language processing
- Mathematics
- Automation

SYLLABUS-AI

- **Introduction to Artificial Intelligence**
- **Intelligent Agents**
- **Problem solving**
- **Knowledge representation and reasoning and Reasoning**
- **Planning and Learning**
- **Applications:** Natural Language Processing(NLP), Expert Systems. RAG

LAB SESSIONS



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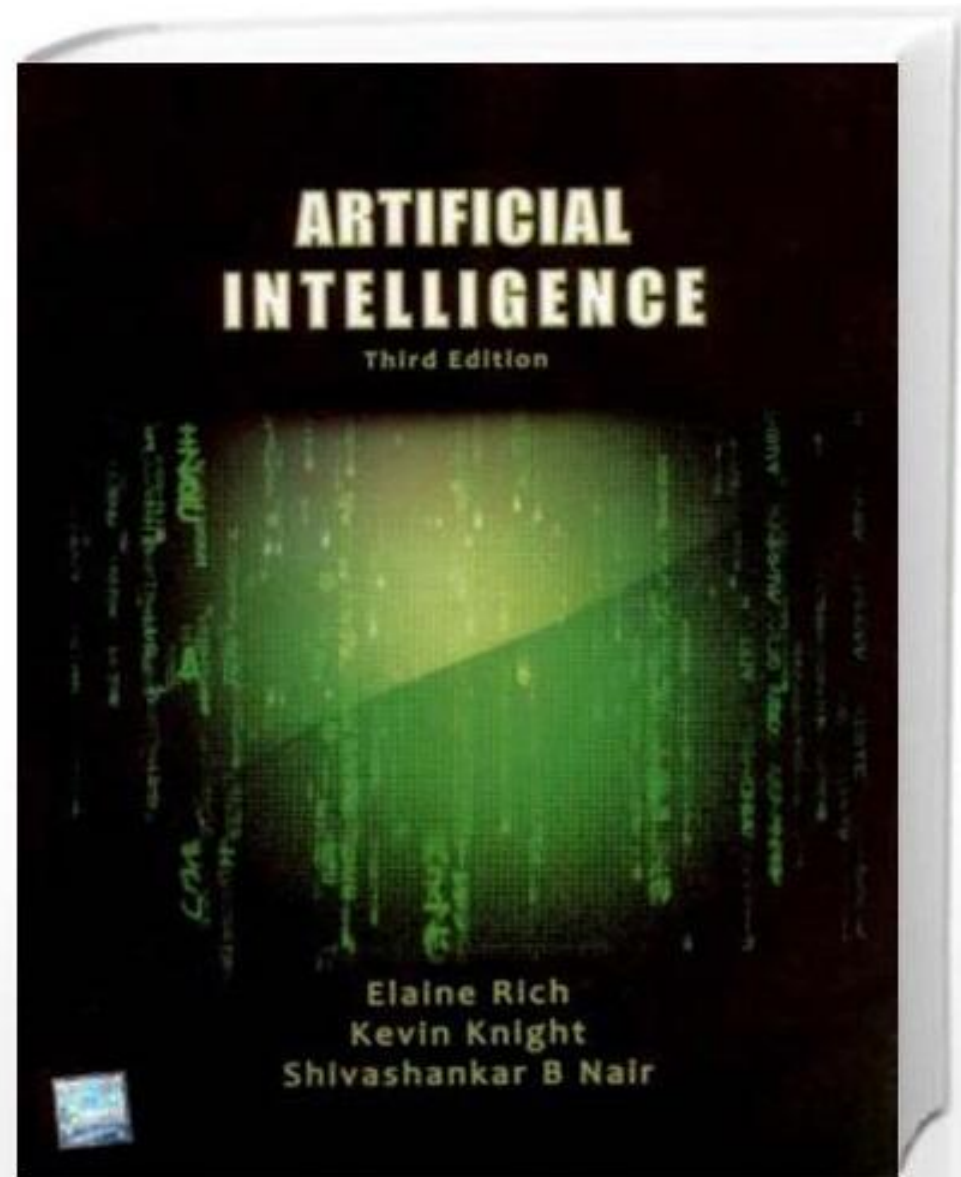
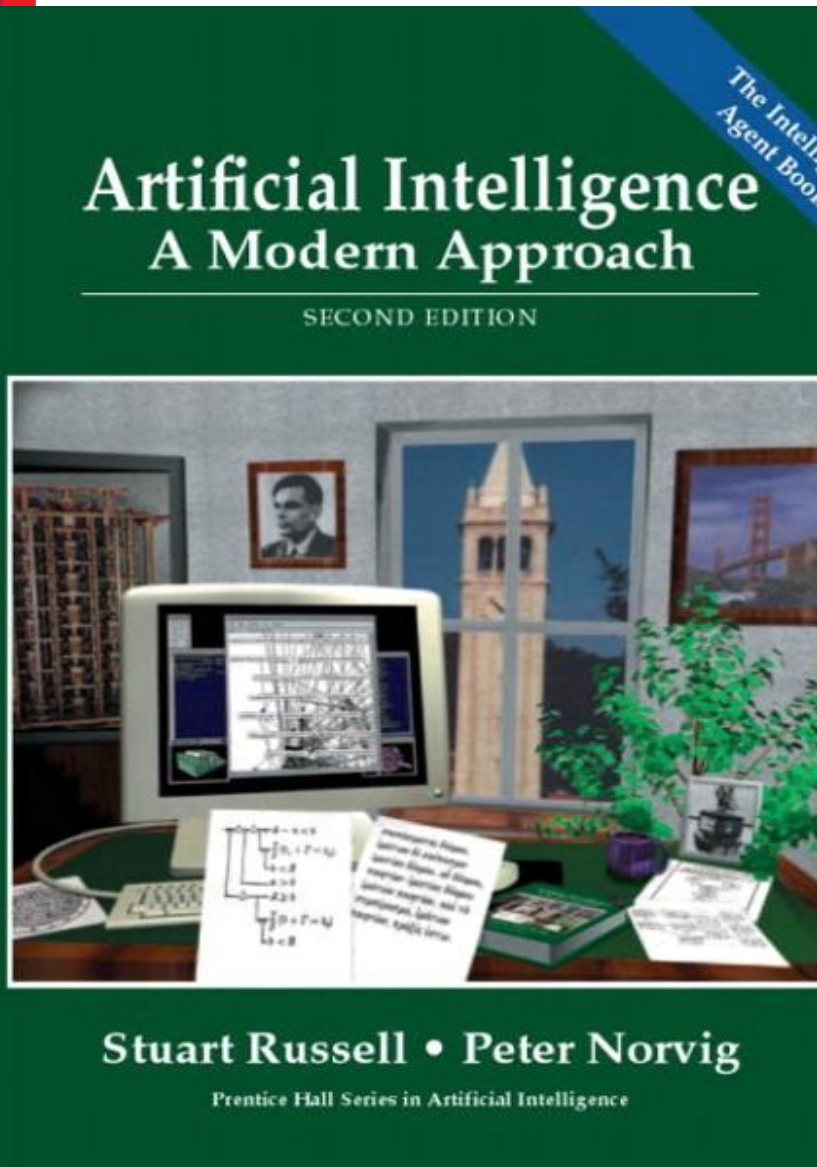


INTERNAL ASSESSMENT

PROGRAMMING LANGUAGES USED FOR AI

- PROLOG
 - PROgramming in LOGic
- JAVA, python

TEXT BOOKS



Teaching Learning methods

- Think-pair-share
- Flipped classroom
- Cooperative-collaborative learning

Limitations of AI

- HIGH COST OF IMPLEMENTATION
- CAN'T REPLACE HUMANS
- DOESN'T IMPROVE WITH EXPERIENCE
- LACKS CREATIVITY unless instructed well
- Hallucinations

Issues with implementation

- Access to Data
- Bias
- Computing Time
- Cost
- Adversarial Attacks
- No Consensus on Safety, Ethics, and Privacy

Limits of AI Today

- Omniscient?
- Clairvoyant?
- General intelligence?
- understanding unconstrained natural language?
- Cannot fully learn a natural language.
- Cannot construct plans in all sorts of dynamic real time domains in general.
- Do not yet exhibit true autonomy and intelligence.

Module 1

Module No.	Unit No.	Details	Hrs.	CO
1		Introduction to Artificial Intelligence	08	CO1
	1.1	History of Artificial Intelligence, Foundations of AI		
	1.2	Categorization of Intelligent System, Components of AI Program,		
	1.3	Sub-areas of AI: Machine Learning, Robotics, Neural Networks, Fuzzy Logic, Cognitive Computing		
	1.4	Applications of AI, Current trends in AI: concepts and applications of: Generative AI, Predictive AI, Discriminative AI, Descriptive AI, Prescriptive AI, Adaptive AI, Interactive AI, Cognitive AI, Explainable AI (XAI), Hybrid AI AI Ethics and Fairness		
		#Self Learning – Exploring and studying Emerging Trends and Research Directions in AI through CASE studies		

WHAT IS AI?



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DEFINITION

- Artificial Intelligence is the study of how to make computers do the things, which at the moment, people do better.
- AI is concerned with designing intelligence in artificial devices.

What is intelligence?

- Behave as intelligently as a human?
- Behave in best possible manner?
- Thinking?
- Acting?

SOME TASK DOMAINS OF AI

MUNDANE TASKS

- Perception
 - Vision
 - Speech
- NLP
 - Understanding
 - Generation
 - Translation
- Commonsense reasoning
- Robot control

SOME TASK DOMAINS OF AI

FORMAL TASKS

- **Games**

- Chess
- Checkers
- Rubic cube

- **Mathematics**

- Geometry
- Logic
- Integral calculus

SOME TASK DOMAINS OF AI EXPERT TASKS

- **Engineering**
 - Design
 - Fault finding
 - Manufacturing planning
- **Scientific analysis**
- **Medical diagnosis**
- **Financial analysis**

AI Foundations

Mathematics

Psychology

Philosophy

Artificial Intelligence

Computer Science

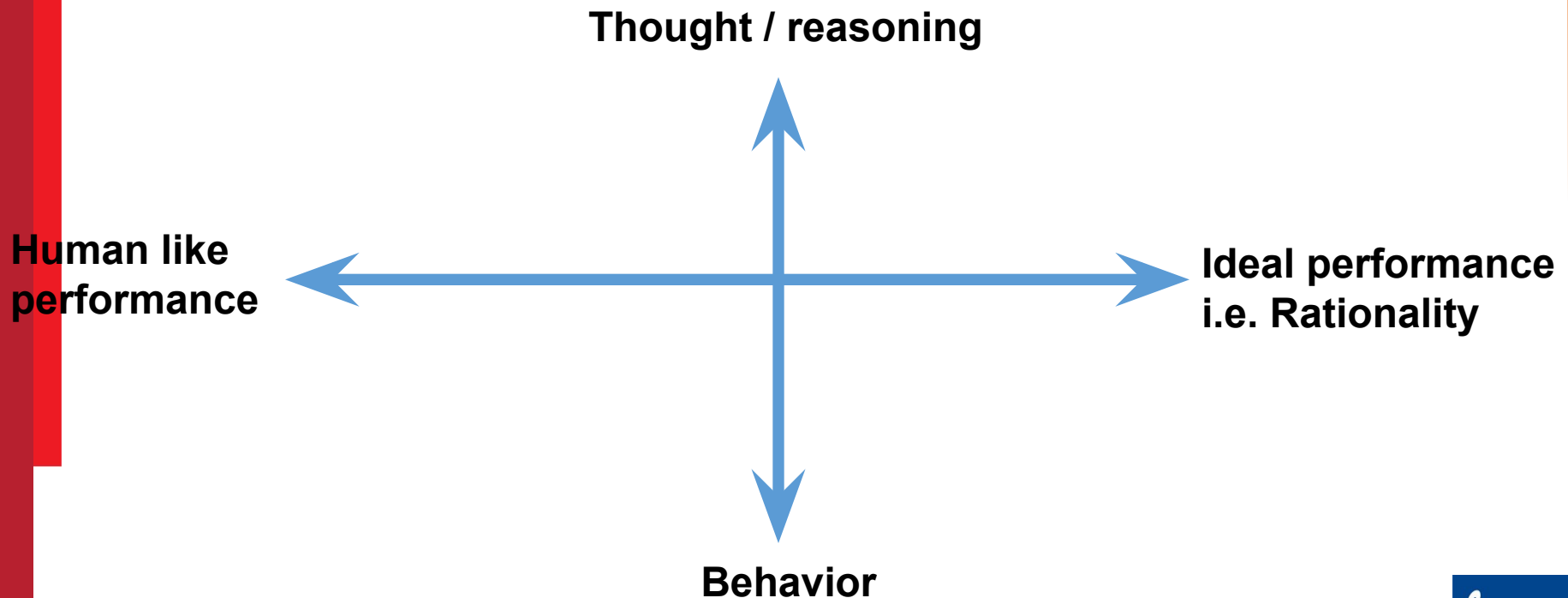
Economics

Biology

Linguistics

AI approaches & Definitions

- Thought process/reasoning Vs behavior
- Human like performance Vs Ideal performance



THEORIES/DEFINITIONS OF AI

1. Systems that think like humans.
2. Systems that think rationally.
3. Systems that act like humans
4. Systems that act rationally

Human-like:

Rationally:

Think:

(1) Cognitive science Approach

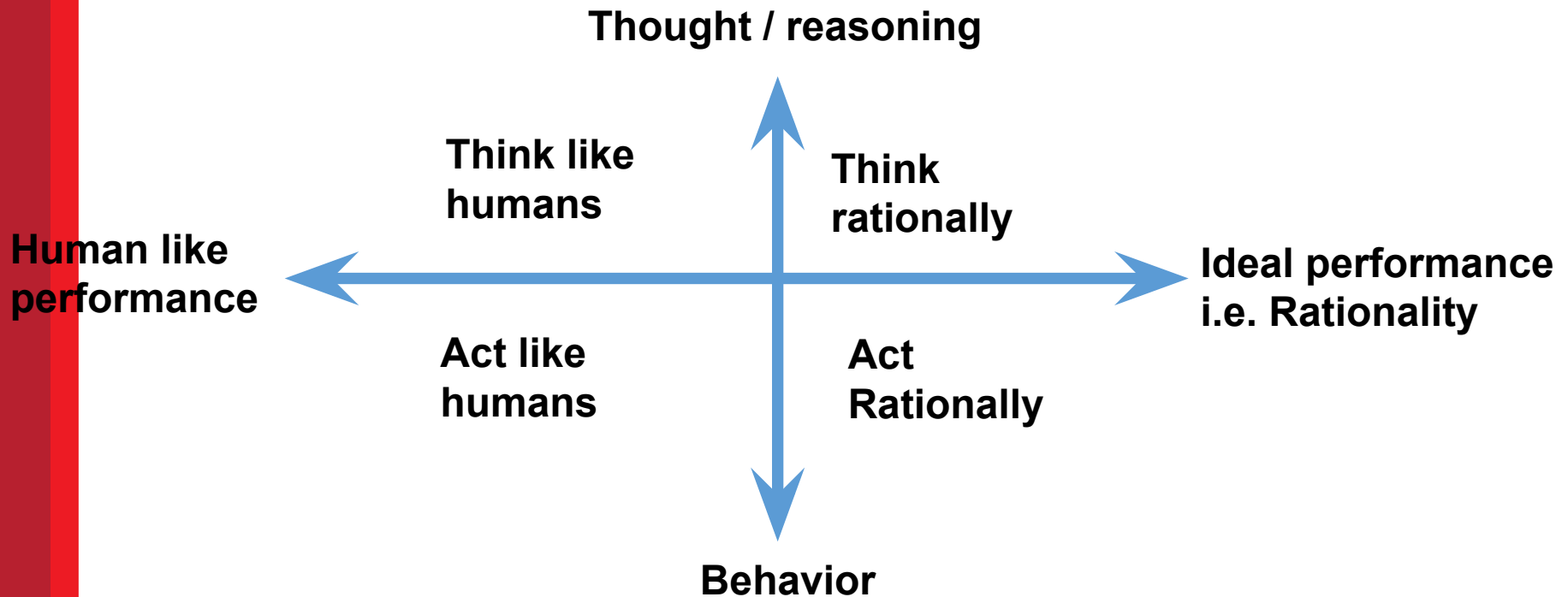
(2) Laws of thought Approach

Act:

(3) Turing Test Approach

(4) Rational agent Approach

AI approaches & Definitions



COGNITIVE SCIENCE APPROACH- THINK LIKE HUMAN

- Automation of activities that we associate with human thinking
- Two ways
 - Through introspection-trying to catch our own thoughts as they go by
 - Through psychological experiments
- Argument: An algorithm performs well on a task and it is therefore a good model of human performance, or vice versa.

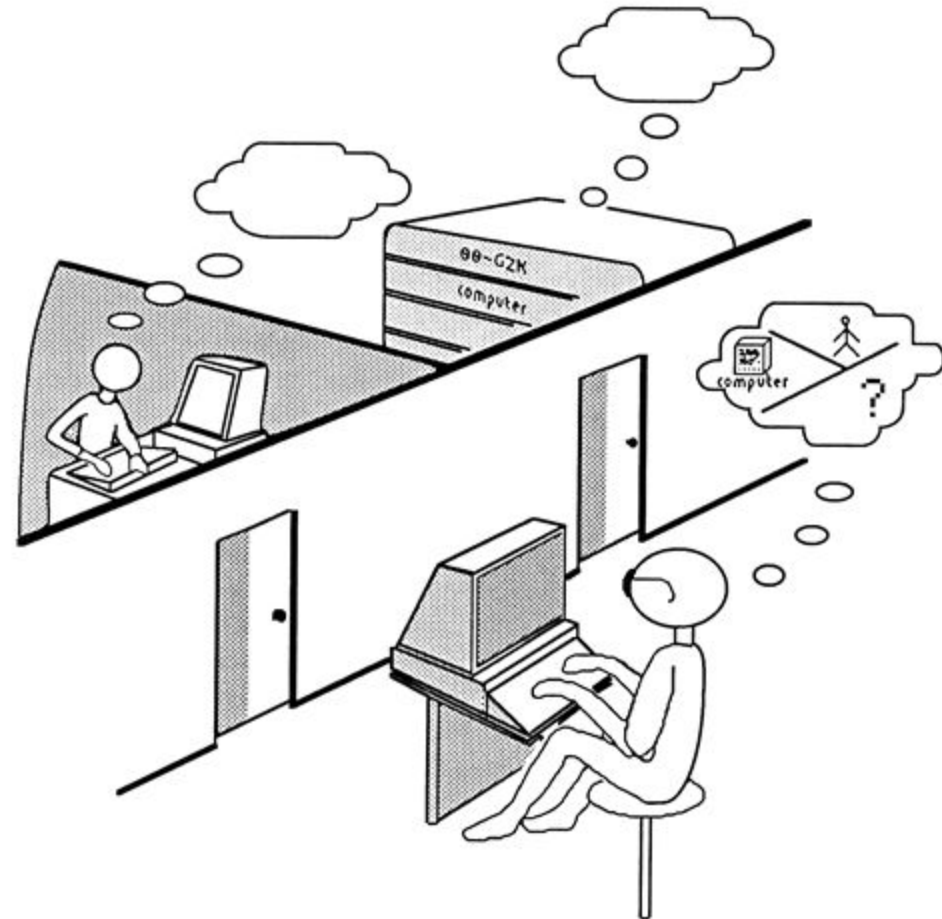
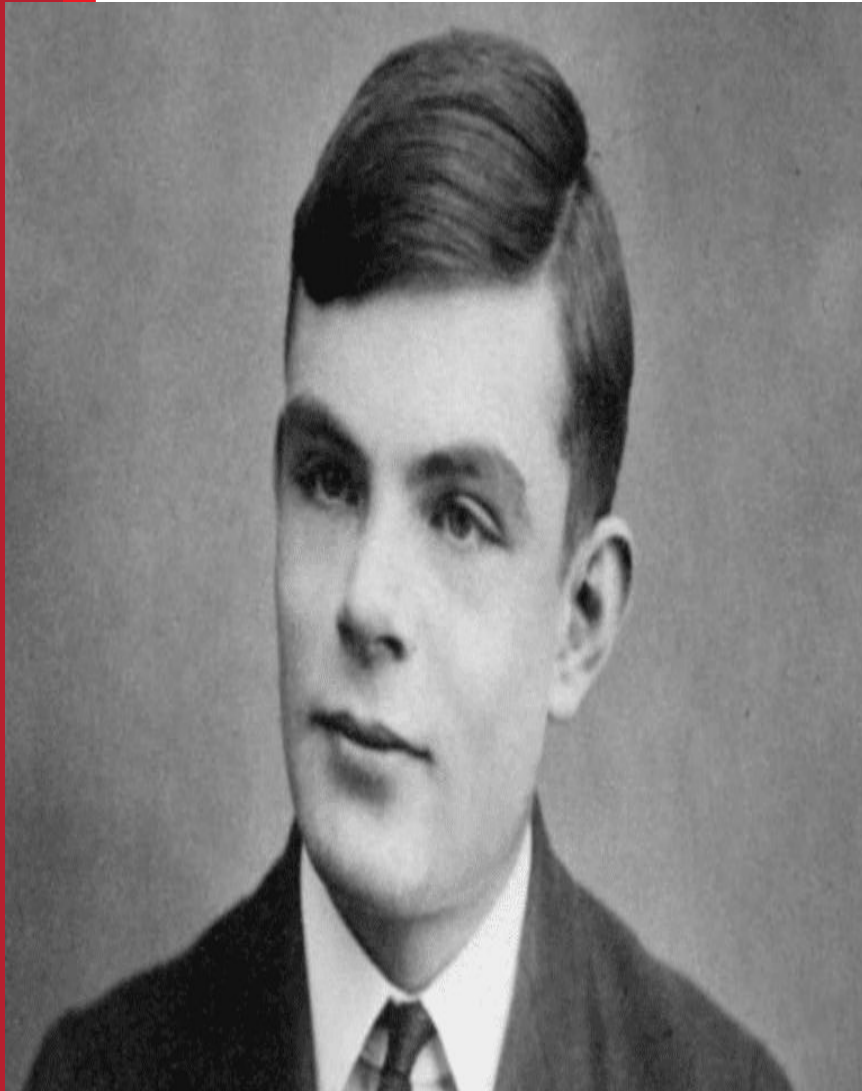
Thinking like human- limitations

- Modern research has separated the two fields of claims: AI and cognitive science.
- The two fields overlap though, especially in the areas of computer vision and natural language.

TURING TEST APPROACH-ACT LIKE HUMAN

- Study of enabling computers to do the things which require human intelligence.
- The machine must have
 - natural language processing
 - knowledge representation
 - automated reasoning
 - machine learning
- Total Turing Test
 - computer vision
 - robotics

Turing Test



Limitations – Acting like human

- Mimicking human behaviour approach is not based on fundamentals on which AI could be based.
- AI developers realized they didn't need to replicate human neural processes or behavioral quirks.
- They could explore broader than just acting like humans
- E.g. studying aerodynamics than building machines those could fly like birds

Laws of thought approach- think Rationally

- Proposed by Aristotle (384-322 B.C.)
- **Syllogisms** provides patterns for argument structures that always yield correct conclusions when given correct premises
- Emphasis is on correct inferences



Limitations – Thinking Rationally

- Problems:
 - Its difficult to represent informal and uncertain knowledge in formal terms required by logical notation
 - There is a big difference between being able to solve a problem "in principle" and doing so in practice (few facts may exhaust all computational power)

Act Rationally- Intelligent agent approach

- Agent and Rational agent
- Involves correct inferences as thinking rationally
- Can suggest actions wherein no correct possible actions can be defined
- Has ability to represent knowledge and reason with it to reach good decisions in a wide variety of situations
- more general than the "laws of thought" approach
- it is more amenable to scientific development than the approaches based on human behaviour or human thought

What has actually been followed?

- Historically, all four approaches to AI have been followed.
- A human-centered approach must be an empirical science, involving hypothesis and experimental confirmation.
- A rationalist approach involves a combination of mathematics and engineering.

Components of AI program

- I/O system – Sensors and Actuators
- Knowledge Representation
 - Facts
 - Rules
- Inference / Reasoning Engine
 - Logic (deduction, induction, abduction)
 - Rule-based reasoning (IF–THEN rules)
 - Probabilistic reasoning (Bayes' theorem)
 - Various algorithms - Search and Problem-Solving Mechanism
- Learning Component
 - Supervised learning (labeled data)
 - Unsupervised learning (pattern discovery)
 - Reinforcement learning (trial-and-error with rewards)
- Control / Agent Architecture
 - Agent model (reactive, deliberative, hybrid)
 - Decision-making strategy
 - Environment interaction loop

Example- Chess playing AI system

Component	Role
Knowledge Representation	Board positions, rules of chess
Inference/Reasoning	Evaluates good vs bad positions
Search	Minimax with Alpha-Beta pruning
Learning	Improves evaluation using past games
Perception	Reads the board state
Action	Makes a move
Control	Manages turn-taking and game flow

Sub Areas of AI

- Machine Learning
- Robotics
- Neural Networks
- Fuzzy Logic
- Cognitive Computing

Sub Areas of AI

- **AI** → the entire field of making machines intelligent
- **ML** → teaches machines from data
- **Neural Networks** → one powerful method used in ML
- **Robotics** → uses AI to make machines act in the physical world
- **Fuzzy Logic** → lets AI reason with uncertainty and vagueness
- **Cognitive Computing** → tries to imitate how humans think, using many AI methods

Machine Learning

- Machine Learning is the branch of AI that enables systems to learn from data and improve performance without being explicitly programmed.
- **Key idea:**
 - Learn patterns from experience.
- **Types:**
 - **Supervised Learning:** learns from labeled data (e.g., spam detection)
 - **Unsupervised Learning:** finds hidden patterns (e.g., clustering customers)
 - **Reinforcement Learning:** learns by trial and error using rewards (e.g., game-playing AI)
- **Applications:**
 - Recommendation systems (Netflix, Amazon)
 - Fraud detection in banking
 - Predictive maintenance in industries

Neural Networks

- Neural Networks are **computational models inspired by the human brain**, consisting of layers of interconnected “neurons” that process information.
- **Key idea:**
 - Learn complex patterns through weighted connections.
- **Types:**
 - ANN (Artificial Neural Networks) – general pattern learning
 - CNN (Convolutional Neural Networks) – images, vision
 - RNN / LSTM – time-series, speech, language
 - Transformers – large language models (like ChatGPT)
- **Applications:**
 - Image recognition
 - Speech recognition
 - Language translation
 - Medical diagnosis
- Neural Networks are a **core technique within Machine Learning**

Robotics

- Robotics is the field of AI that deals with **designing intelligent machines (robots)** that can sense, think, and act in the physical world.
- **Key components:**
 - **Perception:** cameras, sensors
 - **Planning & Control:** deciding actions
 - **Actuation:** motors, movement
 - **AI reasoning:** autonomy and adaptation
- **Applications:**
 - Industrial robots in manufacturing
 - Surgical robots
 - Autonomous vehicles, drones

Fuzzy Logic

- Fuzzy Logic handles **uncertainty and vagueness** by allowing **degrees of truth** instead of strict yes/no (0 or 1).
- **Key idea:**
Truth can be **partial**:
 - Temperature is not just “hot” or “cold” → it can be *somewhat hot*.
- **Applications:**
 - Washing machines, air conditioners
 - Traffic control systems
 - Decision-making systems

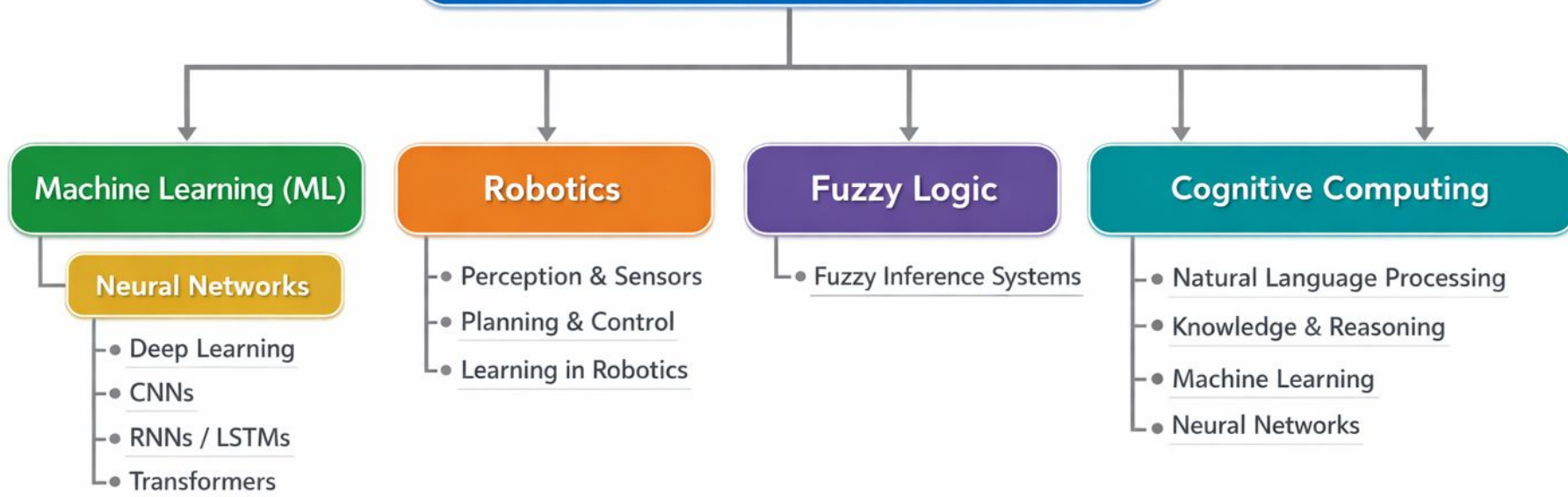
Cognitive Computing

- Cognitive Computing aims to build systems that simulate human thought processes such as understanding language, reasoning, learning, and decision-making.
- **Key characteristics:**
 - Understands **natural language**
 - Learns continuously
 - Reasons with context
 - Handles ambiguity like humans
- **Applications:**
 - Virtual assistants (Siri, Alexa)
 - Medical decision support (IBM Watson)
 - Legal and financial advisory systems
- Cognitive Computing combines multiple AI areas: ML, NLP, neural networks, and knowledge representation.

Sub areas of AI and their Focus

Sub-Area	Main Focus	Relationship to AI
Machine Learning	Learning from data	Core method in modern AI
Neural Networks	Brain-inspired learning models	A technique inside ML
Robotics	Intelligent physical agents	Applies AI in real-world action
Fuzzy Logic	Reasoning with uncertainty	Alternative to strict logic
Cognitive Computing	Human-like reasoning	Integrates many AI techniques

Artificial Intelligence (AI)



Applications of AI



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Current trends in AI and applications

- Generative AI
- Predictive AI
- Discriminative AI
- Descriptive AI
- Prescriptive AI
- Adaptive AI
- Interactive AI
- Cognitive AI
- Explainable AI (XAI)
- Hybrid AI
- AI Ethics and Fairness

PROLOG

FIRST-ORDER Logic Language

Parts of Prolog program

Turbo Prolog

- Domains -optional
- Predicates
- ◉ Clauses
 - Facts
 - Rules
- Goal

Swi Prolog

- Clauses (section heading is not needed)

Writing a prolog program

- Clauses
 - Facts
 - Rules
- Clauses must end with a period.
- Reading a clause
 - Form: Relation/verb(Subject, object,_,).
- example
 - mother(meera, arya). – Meera is mother of Arya
 - now(2025). - now its 2025.
 - gt(25,20).- 25 is greater than 20.
 - type(tom,cat). – Tom's type is cat.

Writing a prolog program

- Facts
 - basic, unchanging relationships or properties in statement
 - form: predicate_name(arguments)
- E.g. person(arya).
 - mother(meera, aya).
 - student(sushil, TY, computer,8.75)
 - distance(mumbai,pune,250)

Writing facts

- Identify relationship in statement – predicate name
- Identify how many arguments are needed to represent information in statement.
- Identify subject, object and other attributes.
- Example
 - man(john).
 - grandfather(john,jen).
 - Item(purse,white).
 - distance(mumbai,pune,250).
 - laptop(hp,8,i7,50000).
- Do not combine multiple conditional statements into one.
 - Wrong: likes(meera,reading),likes(meera,hiking).
 - Correct:
 - likes(meera,reading).
 - likes(meera,hiking).

Writing a prolog program

- Rules

- Represent derived relationships or logic.
- Generally written with variables
- Form: head :- body.
 - where:**Head**: The relationship you're defining.
 - Body**: Conditions under which the relationship is true.

- Example

- `mother(X,Y):- female(X), parent(X,Y).`
- `married(X,Y) :- married(Y,X).`
- `sister(X,Y):-female(X), parent(Z,X), parent(Z,Y).`

Data Types in Prolog

- Variables and their data type declaration isn't mandatory in prolog
- Types:
 - Symbol e.g. used for person, item etc
 - Integer
 - Float
 - List – like arrays
 - String
 - Booleana
 - Free/anonymous varibales

Varibales

- Free variable:
 - not yet bound to a specific value during the execution of a query or rule.
 - It can match with any value, and Prolog may assign it a value during execution to satisfy the conditions of the query or rule.
 - Same variables can be used for different rules in same program.
- Starts with an uppercase letter or an underscore (e.g., X, Person, _Variable).
- Acts as a placeholder that Prolog will attempt to bind to values to satisfy goals.
- If left unbound, Prolog considers it as "free" and may include it in the result of a query.

Free variable example

- Example
female(meera).
female(arya).
mother(meera,arya).

Query: female(X)

o/p : X=meera, X=arya

Query: mother(X,arya)

o/p: X=meera

Anonymous variable

- Always represented by an underscore (`_`)
- is used when a value is needed for a condition but the specific value is irrelevant.
- Useful for ignoring parts of a predicate or argument that are not needed for the solution
- Example-
 - `long_novel(Title) :- written_by(_, Title), book(Title, Length), Length > 300.`

Turbo Prolog program

domains

name, sport=symbol

predicates

likes(name,sport)

clauses

likes(ellen, tennis).

likes(john, football).

likes(tom, baseball).

likes(eric, swimming).

likes(mark, tennis).

likes(bill, Activity) if likes(tom,
Activity).

Swi Prolog program

likes(ellen, reading).

likes(john, computers).

likes(john, badminton).

likes(leonard, badminton).

likes(eric, swimming).

likes(eric, reading).

likes(bill, Activity) if
likes(tom, Activity).

domains

name, sport=symbol

predicates

likes(name, sport)

clauses

likes(ellen, tennis).

likes(john, football).

likes(tom, baseball).

likes(eric, swimming).

likes(mark, tennis).

likes(bill, Activity) if likes(tom, Activity).

Rules

- Difference between predicates and clauses (For turbo prolog)
 - likes(person, sports)
 - likes(ram, tennis).
- Capitalization and free variables
- Grouping of predicates/clauses
- No nesting of predicates
 - Father(dashrath,father(kush)).

Rules – capitalizations

- Do not capitalize first letters of arguments as they would be taken as variables.
- If you wish to write in capital and wish it to be treated as constant, use quotes- “” as:
 - `phone_number("Albert", "EZY-3665")`.

Rules - Grouping

- Clauses/Statements of same predicate should be grouped together.
- Example-
 - Write all facts and rules of person, then write all facts and rules of mother, then write all facts and rules of father etc.
 - Do not write one fact of person, then one of mother, then one of father, then again fact of person/mother/father etc.

SAMPLE OUTPUT

```
———— Dialog ————  
Goal: likes<ellen,X>  
X=tennis  
1 Solution  
Goal: likes<P,tennis>  
P=ellen  
P=mark  
2 Solutions  
Goal: likes<tom,baseball>  
True  
Goal: S
```

domains

product, sum = integer

predicates

add_em_up(sum, sum, sum)

multiply_em(product, product, product)

clauses

add_em_up(X, Y, Sum) :- Sum = X + Y.

multiply_em(X, Y, Product) :- Product = X *
Y.

DATABASE VS KNOWLEDGEBASE

- Rule base Vs knowledgebase programming (C/Java Vs Prolog/LISP)

	Rule Based System	Knowledge Based System
Can Process	Data Rules	Data Rules Knowledge
Can Output	Information Decision	Information, Decision, Expert Advice, Recommendations
Commonly used for	Enterprise rules	Domain specific rules
Ideal for	Simplistic business rules	Complex business rules
Best for these types of applications	Decision Compliance	Advising, Product Selection, Recommending, Troubleshooting
Domain	Broad Logic	Deep Logic

Queries?

Thank you!!